CHAPTER 06

Aldehydes and Ketones I

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 78a84ed39319352c38d2b4333197d617eec4594e7d4ad5761c5aa4e137195f91

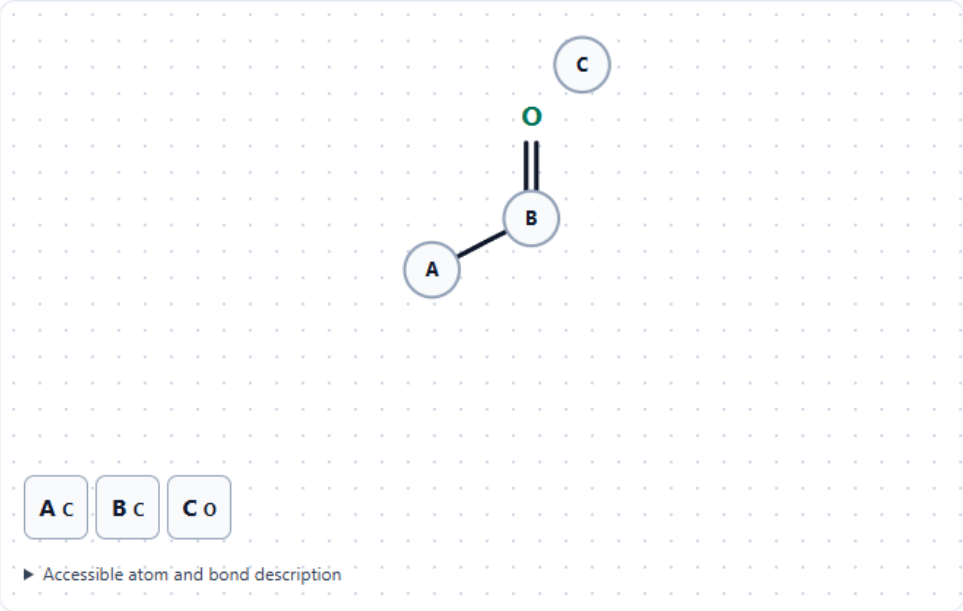
BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

addition-site | Independent exercise

The canvas retains the starting ethanal to trace one electron-flow step. The incoming nucleophile and tetrahedral product are not drawn.

Screenshot: addition-site-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=189.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

addition-site:stage-1 | Select the atom attacked by a carbon nucleophile.

Answer type: atoms

addition-site:stage-2 | As the new sigma bond forms, the C=O pi pair moves to...

Answer type: choice

Option 0 (A): Oxygen

Option 1 (B): The incoming nucleophile

Option 2 (C): A methyl hydrogen

addition-site:stage-3 | The immediate former-carbonyl geometry becomes approximately...

Answer type: choice

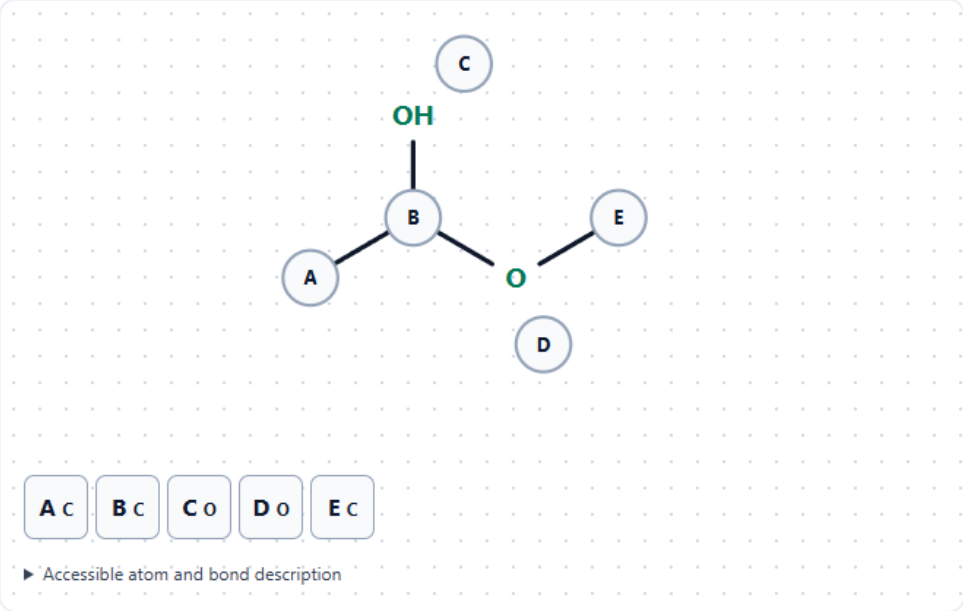
Option 0 (A): Tetrahedral

Option 1 (B): Linear

Option 2 (C): Unchanged trigonal planar

hemiacetal-identify | Independent exercise

Screenshot: hemiacetal-identify-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=120.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=120.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=180.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=240.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [3, 4, 1]]

hemiacetal-identify:stage-1 | Select the carbon bearing two oxygen substituents.

Answer type: atoms

hemiacetal-identify:stage-2 | One OH and one OR on that carbon identify a…

Answer type: choice

Option 0 (A): Hemiacetal

Option 1 (B): Acetal

Option 2 (C): Geminal diol

hemiacetal-identify:stage-3 | Converting it to an acetal with alcohol is favored by…

Answer type: choice

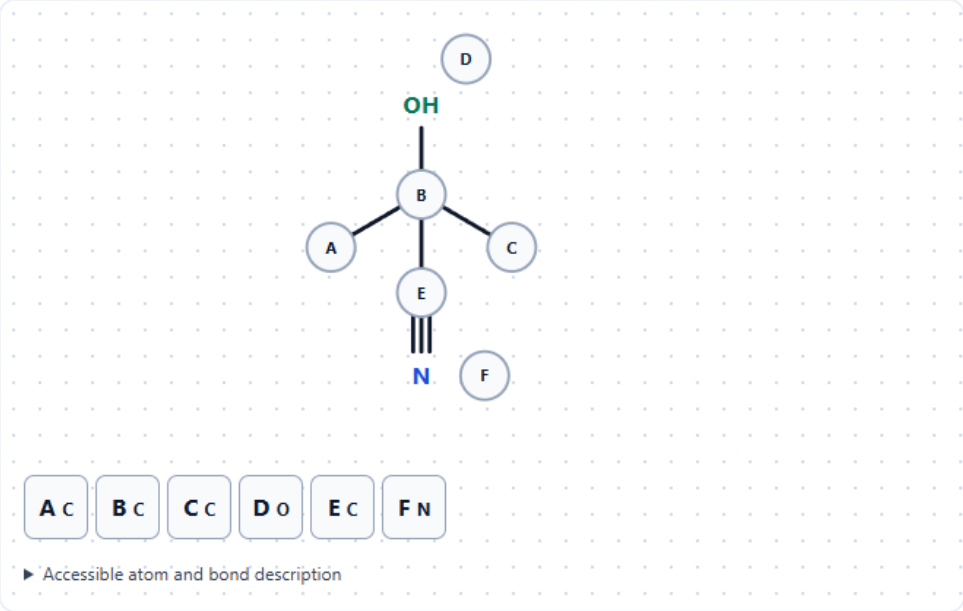
Option 0 (A): Acid catalysis and removal of water

Option 1 (B): Adding abundant water alone

Option 2 (C): Strong base alone

cyanohydrin-bond | Independent exercise

Screenshot: cyanohydrin-bond-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=120.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=120.00; charge=0; explicit H override=none

3 (D): O; x=120.00, y=25.00; charge=0; explicit H override=none

4 (E): C; x=120.00, y=150.00; charge=0; explicit H override=none

5 (F): N; x=120.00, y=205.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1], [4, 5, 3]]

cyanohydrin-bond:stage-1 | Select the nitrile carbon added by cyanide.

Answer type: atoms

cyanohydrin-bond:stage-2 | How many carbons are in the cyanohydrin?

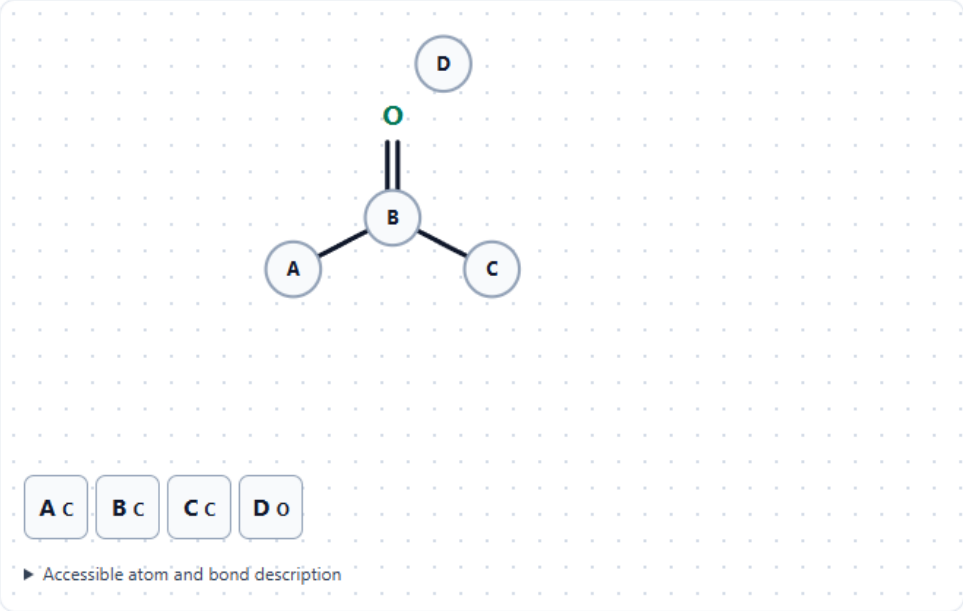
Answer type: number | unit: carbons

cyanohydrin-bond:stage-3 | Select both carbon endpoints of the new C–C bond formed by cyanide addition.

Answer type: atoms

hydride-product | Independent exercise

Screenshot: hydride-product-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

hydride-product:stage-1 | Select the carbonyl carbon receiving hydride.

Answer type: atoms

hydride-product:stage-2 | After protonation, the product is a…

Answer type: choice

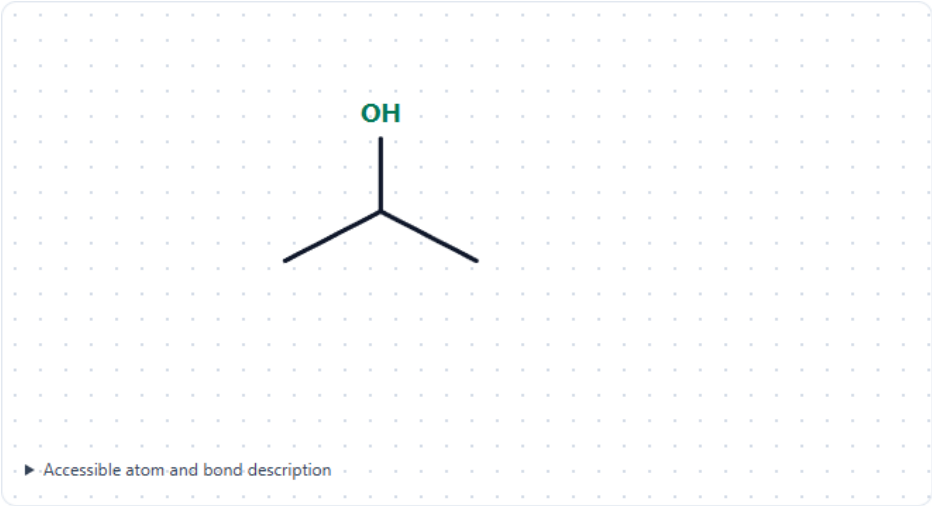
Option 0 (A): Secondary alcohol

Option 1 (B): Primary alcohol

Option 2 (C): Tertiary alcohol

This stage uses a different molecule (connectivity shown).

Screenshot: hydride-product-stage-3-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

hydride-product:stage-3 | Name the product.

Answer type: text